

**Assignment Submission Sheet**

**ARIN 340**

AI Applications

Unit 4 Assignment: Exploring Content Creation Using Generative AI

***Matthew Kolonich***

***Dr. Holbert***

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**Tool Exploration:**

For this assignment, I chose six different AI tools across three areas: text, images, and sound. I wanted to see how each tool could contribute to a themed project about marine life, and together they created a complete learning experience that blended storytelling, visuals, and audio.

I first used ChatGPT and Claude for text input. I chose ChatGPT because it is great at coming up with creative ideas and writes in a way that is fun and engaging. It helped me write a children's story about a sea turtle named Tali who explores the reef and learns about protecting the ocean. ChatGPT made it easy to include dialogue, sensory details, and even repeating rhymes that fit the theme of a read-aloud story. I picked Claude because it is good at structure and clarity. While ChatGPT gave me imagination, Claude gave me organization. I used Claude to make a classroom lesson plan and worksheet to go with the story, which included learning objectives, activities, and an assessment rubric. These two text tools balanced creativity with structure.

For my next 2 tools I worked with DALL·E 3 and Adobe Firefly to work with images. I chose DALL·E 3 because it can create bright, colorful illustrations that look like they belong in a children's book. This was perfect for making a poster of Tali the turtle and her friends in the reef. Adobe Firefly, on the other hand, is better for clean, simple designs. I used it to make social media graphics with bold text and icons. I picked Firefly because it also has strong accessibility features, like good color contrast, which makes the images easier for everyone to read.

For sound, I tried ElevenLabs and Suno. ElevenLabs is one of the best tools for turning text into realistic speech, so I used it to narrate the story. I liked that I could adjust the speed and emphasis to make it sound right for children. Suno was a good match because it creates background music and soundscapes. I used it to generate ocean sounds like waves and whale calls, which added atmosphere and made the story feel more immersive.

In the end, these six tools worked together as a multimedia toolkit. ChatGPT and Claude gave me words, DALL·E 3 and Adobe Firefly gave me visuals, and ElevenLabs and Suno gave me sound. I chose each one because of its strengths, and together they helped me build a project about marine life that feels complete and engaging.



**Prompt Iteration:**

- Write a summary (1-2 pages) of your AI tool selection, including:
  - Within each generative AI tool (six initial prompts total, two within three content creation categories), state your initial prompt and your successive prompts (three prompts minimum, per tool, including the initial prompt).
  - Analyze the prompt iteration process for each generative AI tool. There should be one paragraph or several sentences for each generative AI Tools

For the project, I used six generative AI tools across text, images, and audio. Each tool required several prompt iterations to get the results that I was looking for. Below is a summary of how I refined my prompts and what I learned from each tool.

**ChatGPT (text – children’s story)**

I started by using text with ChatGPT. I started by asking for a children’s story about a sea turtle named Tali. The first draft was creative but needed more detail for a more storybook feel. I adjusted my prompts to add illustration for each page and then refined it further by asking for a repeating rhyming chorus. . The process showed me how small changes in prompts, like asking for rhyme or specific word count, can shift a story from good to great. Another big aspect I noticed was that the illustrations were in text. ChatGPT gave me the ideas for the illustrations but not the actual illustrations. I stopped at the 3<sup>rd</sup> iteration because it had created the storybook with rhyme. I stopped after my 2nd iteration because the story was detailed, organized, and had created cues for the pictures.

**Claude (text – lesson plan)**

I next tried text with Claude. I wanted to create a lesson plan for the storybook. My first prompt produced an outline, but I realized it needed more a details and plans for a real life classroom. In my 2nd iteration, I added accessibility considerations and aligned the plan with NGSS standards. NGSS standards are learning standards for elementary students. Finally, I asked for a worksheet with questions and an action plan template. The prompt iterations made the lesson easier to use. This process showed me that Claude works best when given detailed structure, since it responds well to organized, step-by-step requests. I stopped because it had created the lesson plan and had a worksheet for students.

**DALL·E 3 (images – poster)**

When I used DALL·E 3, my first prompt gave me a bright poster of Tali and the reef, but the text in the image came out distorted. In my second prompt, I asked for no text and cleaner coral textures, which fixed the issue. My last iteration focused on resolution and color palette for print quality. This shows that DALL·E is great for visuals but struggles with in-image text, so it’s better to add text later with design software. Prompt refinements worked best when they targeted visual issues like color, detail, or resolution. I stopped here because I tried to refine the picture more and all further images looked almost exactly the same.

**Adobe Firefly (images – social graphics)**

With Adobe Firefly I want to make a simple, shareable graphic. The first attempt produced decent tiles, but some text and icons were hard to read. I improved this by asking for stronger contrast and simplified icons, which worked well. My last prompt added background options and brand space for logos. The process showed that Firefly is strong at accessibility and layout design, but the prompts need to be

specific about readability and style. Iterations here were more about refining usability rather than fixing big flaws. I stopped here because why the color and background changed, most of the main design stayed the same.

#### ElevenLabs (audio – narration)

For narration, I used ElevenLabs to read the story aloud. My first attempt sounded natural, but the pacing was too fast for me. I adjusted the prompts to slow down the reading and add pauses using SSML (Speech Synthesis Markup Language). In the final version, I created both a clean narration and one mixed with subtle ocean background. I learned that ElevenLabs is powerful but requires fine-tuning with prompts to control pacing, emphasis, and tone. Without those refinements, the narration can feel rushed or uneven. I stopped at this point because the speed of the reading seemed well. ElevenLabs read each version of the story as I had asked, but I could not find out how to download an output of the audio so I have not included the results.

#### Suno (audio – ambient sounds)

I next tried Suno. I wanted a calm ocean background. The first soundscape was nice but had too many distracting bell sounds. The bells and chimes distracted in the background. My next prompt reduced those and balanced the audio so it wouldn't compete with the narration. In my final prompt, I asked for separate audio stems for waves, whales, and pads, which gave me more flexibility when mixing. This showed me that Suno is great for generating ambience but works best when prompts break down the layers of sound. Reiterating the prompts helped to make the soundscape sound more clear and relaxing. It also slowed the speed of the music. I wasn't able to download the finished soundscapes, but I do have the prompts I used.

### ChatGPT-Text-Create a Storybook for Children

**Initial Prompt-Write a 600-word children's story (ages 6–9) about a curious young sea turtle named Tali who explores a coral reef, meets friends (a clownfish, an octopus, and a wise old manta ray), and learns one simple thing each character does to protect the ocean. Use vivid sensory language, short paragraphs, friendly dialogue, and end with a warm, memorable moral line. Keep the tone hopeful and adventurous.**

#### Tali's Big Coral Reef Adventure

Tali was a little green sea turtle with bright eyes and a heart full of questions. Every morning she poked her head above the water and wondered, *What's waiting for me today?*

One sunny morning, the ocean shimmered like liquid glass. The coral reef stretched before her in glowing colors—red fans, golden sponges, and purple towers. Tiny fish darted between the coral like sparks of rainbow light.

Tali paddled her flippers eagerly. "I want to explore every corner!" she said aloud, bubbles popping from her beak.

As she swam, she noticed a small orange clownfish weaving in and out of a sea anemone's swaying tentacles.

"Hello!" Tali called.

The clownfish peeked out shyly. "Hi, I'm Cora."

"I'm Tali. What are you doing?"

"I'm keeping the anemone clean," said Cora proudly. "I nibble away the bits of leftover food. It keeps my home healthy and gives me protection, too."

Tali's eyes widened. "So by cleaning, you're helping the ocean?"

Cora nodded. "Every little action matters."

"Wow," Tali said, and with a flick of her flippers she waved goodbye and swam on.

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Farther along, the reef grew quiet and shadowy. Tali noticed something shift in a crack between rocks. Out slithered an octopus, her skin changing from sandy brown to soft pink.

"Who's there?" asked the octopus in a gentle, gurgly voice.

"It's just me, Tali!"

The octopus smiled and stretched out an arm. "I'm Ollie. What brings you to my part of the reef?"

"I'm exploring! Do you do anything special to help the ocean?"

Ollie chuckled and puffed a tiny cloud of water. "Of course. I love hiding treasures, but I never leave trash behind. If I find a shiny can or a lost bottle, I tuck it away safely in the sand or near the rocks. That way no fish get hurt."

Tali clapped her flippers together. "That's clever!"

Ollie winked. "We all do our part, big or small."

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Tali swam onward until the reef opened wide into a blue underwater meadow. The water grew deeper, darker, calmer. Suddenly, a shadow glided above her like a gentle cloud.

"Hello down there," boomed a slow, kind voice.

Tali craned her neck. A manta ray soared overhead, wings spreading as wide as a boat. His eyes twinkled kindly.

"Wow!" Tali gasped. "You're enormous!"

The manta ray chuckled. "I'm Mako. Don't be afraid—I'm a friend."

Tali swam alongside him. "Mako, how do *you* help the ocean?"

"I filter tiny plankton from the water as I swim," Mako explained. "By eating the smallest bits of food, I keep the water clear and balanced. The ocean needs all creatures, big and small, to do their part."

Tali gazed at the manta ray's graceful movements. "That's beautiful."

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As the sun began to set, golden light spilled down through the water. Tali paddled back toward her favorite coral rock, her mind buzzing with everything she'd learned.

Cora cleaned her anemone home.

Ollie tucked away dangerous trash.

Mako filtered the sea to keep it clear.

Tali rested on her rock and whispered to herself, "If they can help, so can I."

She decided she would never nibble at floating plastic, and whenever she saw seaweed tangling on coral, she'd tug it free.

The reef glittered in the fading light, alive with color and song. Tali felt proud, small but strong, like a tiny shell in a vast, shining ocean.

She curled her flippers close and thought, *The ocean is our home, and every friend makes it better.*

And that night, beneath the silver moonlight, Tali drifted to sleep with a smile, carrying a new truth in her heart:

When each of us cares in small ways, together we keep the ocean safe and beautiful.

**First Iteration- Same story but include 6 short illustrations(one per page spread) — add very brief captions like “Page 1: Tali peeks into reef, sunbeams, sea grass (wide shot)”.**

### **Tali’s Big Coral Reef Adventure**

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#### **Page 1: Tali peeks into reef, sunbeams, sea grass (wide shot)**

Tali was a little green sea turtle with bright eyes and a heart full of questions. Every morning she poked her head above the water and wondered, *What’s waiting for me today?*

One sunny morning, the ocean shimmered like liquid glass. The coral reef stretched before her in glowing colors—red fans, golden sponges, and purple towers. Tiny fish darted between the coral like sparks of rainbow light.

Tali paddled her flippers eagerly. “I want to explore every corner!” she said aloud, bubbles popping from her beak.

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#### **Page 2: Cora the clownfish in orange anemone, Tali watching with wonder**

As she swam, she noticed a small orange clownfish weaving in and out of a sea anemone’s swaying tentacles.

“Hello!” Tali called.

The clownfish peeked out shyly. “Hi, I’m Cora.”

“I’m Tali. What are you doing?”

“I’m keeping the anemone clean,” said Cora proudly. “I nibble away the bits of leftover food. It keeps my home healthy and gives me protection, too.”

Tali’s eyes widened. “So by cleaning, you’re helping the ocean?”

Cora nodded. “Every little action matters.”

“Wow,” Tali said, and with a flick of her flippers she waved goodbye and swam on.

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#### **Page 3: Ollie the octopus stretching an arm, colorful coral behind, Tali leaning closer**

Farther along, the reef grew quiet and shadowy. Tali noticed something shift in a crack between rocks. Out slithered an octopus, her skin changing from sandy brown to soft pink.

“Who’s there?” asked the octopus in a gentle, gurgly voice.

“It’s just me, Tali!”

The octopus smiled and stretched out an arm. “I’m Ollie. What brings you to my part of the reef?”

“I’m exploring! Do you do anything special to help the ocean?”

Ollie chuckled and puffed a tiny cloud of water. “Of course. I love hiding treasures, but I never leave trash behind. If I find a shiny can or a lost bottle, I tuck it away safely in the sand or near the rocks. That way no fish get hurt.”

Tali clapped her flippers together. “That’s clever!”

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**Page 4: Enormous manta ray gliding above Tali in deep blue water, sunlight filtering down**

Tali swam onward until the reef opened wide into a blue underwater meadow. The water grew deeper, darker, calmer. Suddenly, a shadow glided above her like a gentle cloud.

“Hello down there,” boomed a slow, kind voice.

Tali craned her neck. A manta ray soared overhead, wings spreading as wide as a boat. His eyes twinkled kindly.

“Wow!” Tali gasped. “You’re enormous!”

The manta ray chuckled. “I’m Mako. Don’t be afraid—I’m a friend.”

Tali swam alongside him. “Mako, how do *you* help the ocean?”

“I filter tiny plankton from the water as I swim,” Mako explained. “By eating the smallest bits of food, I keep the water clear and balanced. The ocean needs all creatures, big and small, to do their part.”

Tali gazed at the manta ray’s graceful movements. “That’s beautiful.”

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**Page 5: Golden sunset light glowing through reef, Tali resting on a rock, Cora, Ollie, and Mako nearby**

As the sun began to set, golden light spilled down through the water. Tali paddled back toward her favorite coral rock, her mind buzzing with everything she’d learned.

Cora cleaned her anemone home.

Ollie tucked away dangerous trash.

Mako filtered the sea to keep it clear.

Tali rested on her rock and whispered to herself, “If they can help, so can I.”

She decided she would never nibble at floating plastic, and whenever she saw seaweed tangling on coral, she’d tug it free.

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**Page 6: Starry moonlight shining over reef, Tali curled up peacefully, fish sleeping nearby**

The reef glittered in the fading light, alive with color and song. Tali felt proud, small but strong, like a tiny shell in a vast, shining ocean.

She curled her flippers close and thought, *The ocean is our home, and every friend makes it better.*

And that night, beneath the silver moonlight, Tali drifted to sleep with a smile, carrying a new truth in her heart:

When each of us cares in small ways, together we keep the ocean safe and beautiful.

**2<sup>nd</sup> Iteration- Keep the story but add a 4-line rhyming chorus that repeats after each reef encounter. Keep rhymes simple and singable, suitable for read-aloud.**

**Tali’s Big Coral Reef Adventure**



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**Page 1: Tali peeks into reef, sunbeams, sea grass (wide shot)**

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**Chorus:**

*Care for the ocean, clean and bright,*

*Little deeds can make things right.*

*Friends together, big and small,*

*Helping the sea is good for all.*

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**Page 3: Ollie the octopus stretching an arm, colorful coral behind, Tali leaning closer**

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**Claude-Text-Create A Lesson Plan**

**Initial Prompt-Create a 1-page lesson plan (grade 4–6) titled “Reef Guardians”. 45-minute lesson, learning objectives (3), materials list, step-by-step activities (including a short read-aloud of a story about a turtle named Tali), one hands-on activity, assessment rubric (3 criteria), and 3 extension/homework ideas.**

#### Reef Guardians

Grade Level: 4-6 | Duration: 45 minutes | Subject: Environmental Science

#### Learning Objectives

By the end of this lesson, students will be able to:

1. Identify at least 5 different organisms that live in coral reef ecosystems and explain their roles
2. Describe 3 major threats facing coral reefs and their impact on marine life
3. Create an action plan with specific steps they can take to help protect coral reefs

#### Materials List

- Read-aloud story: "Tali's Reef Adventure" (provided below)
- Colored pencils/markers
- Large paper (11x17" or poster board)
- Reef ecosystem fact cards
- Glue sticks
- Sample images of coral reef animals
- Assessment rubric handouts

#### Step-by-Step Activities

##### Opening Hook (5 minutes)

Show students a vibrant image of a coral reef. Ask: "What do you notice? What questions do you have?" Record responses on the board.

##### Read-Aloud: Tali's Reef Adventure (10 minutes)

*"Tali the sea turtle glided gracefully through the crystal-clear waters of her coral reef home. The reef bustled with life – clownfish darted between sea anemones, parrotfish crunched on coral, and cleaner wrasse set up their underwater 'car wash' stations. But lately, Tali noticed troubling changes. Some corals had turned ghostly white, plastic bags drifted like jellyfish, and fewer fish swam among the branches. 'I must find a way to help my reef family,' Tali thought, determined to become a true reef guardian. She began spreading the word to other sea creatures and even the humans who visited, teaching them how small actions could make a big difference in protecting their underwater paradise."*

Discuss: What did Tali notice? How do you think she felt? What might she do to help?

##### Reef Ecosystem Exploration (15 minutes)

1. Divide class into groups of 3-4 students
2. Give each group reef ecosystem fact cards covering:

- Coral polyps and zooxanthellae partnership
  - Fish species (clownfish, parrotfish, angelfish)
  - Other creatures (sea turtles, sharks, octopi)
  - Threats (pollution, warming waters, overfishing)
3. Groups rotate through stations, reading cards and taking notes
  4. Quick share-out: Each group shares one fascinating fact they learned

Hands-On Activity: Design a Reef Protection Campaign (12 minutes)

Students work in pairs to create a mini-poster campaign featuring:

- Title: Creative slogan for reef protection
- Problem: One major threat to reefs (with illustration)
- Solution: Two specific actions people can take
- Call to Action: Encouraging message for others

Encourage creativity with drawings, colors, and persuasive language.

Closing Circle (3 minutes)

Students share one action from their poster that they personally commit to trying. Create a class "Reef Guardian Pledge" list.

Assessment Rubric

| Criteria                | Excellent (4)                                                                           | Proficient (3)                                                | Developing (2)                                        | Beginning (1)                                               |
|-------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| Ecosystem Understanding | Identifies 5+ reef organisms and clearly explains their relationships and roles         | Identifies 4-5 organisms with good explanations of most roles | Identifies 3-4 organisms with basic explanations      | Identifies fewer than 3 organisms with unclear explanations |
| Threat Awareness        | Describes 3+ threats with detailed impact explanations and connections to real examples | Describes 3 threats with good impact explanations             | Describes 2-3 threats with basic impact understanding | Describes 1-2 threats with minimal understanding            |
| Action Planning         | Creates detailed, realistic action plan with 4+ specific, achievable steps              | Creates solid action plan with 3-4 specific steps             | Creates basic action plan with 2-3 general steps      | Creates minimal action plan with 1-2 vague steps            |

Extension Activities & Homework

Option 1: Reef Research Project

Choose one coral reef animal (not discussed in class) and create a fact file including: habitat, diet, reef role, threats it faces, and one surprising fact. Present findings in a creative format (poster, comic strip, or short video).

Option 2: Home Ocean Impact Audit

Conduct a week-long family audit of ocean-impacting choices: plastic use, seafood consumption, sunscreen types, and water usage. Create a before/after comparison chart and propose three family changes to be more reef-friendly.

Option 3: Community Action Letter

Write a persuasive letter to a local business, school administrator, or government official explaining coral reef threats and requesting specific action (like reducing plastic use, supporting reef conservation, or educational initiatives). Include facts from the lesson and personal reasons why reef protection matters.

I've created a comprehensive "Reef Guardians" lesson plan that meets all your requirements! The lesson is designed to be engaging and educational, featuring:

**Key Highlights:**

- **Tali's story** introduces students to reef ecosystems through a relatable sea turtle character who notices environmental changes
- **Hands-on campaign creation** lets students apply their learning by designing reef protection posters
- **Progressive learning** moves from exploration to understanding to action
- **Differentiated extensions** offer three different homework options to match various learning styles and interests

The lesson balances scientific content with environmental stewardship, encouraging students to see themselves as capable protectors of ocean ecosystems. The assessment rubric focuses on understanding, awareness, and actionable planning rather than just memorization.

Would you like me to modify any section or add additional detail to specific activities?

**1<sup>st</sup> iteration- Add accessibility modifications for diverse learners (ELL, visual impairment) and align the lesson to one NGSS standard (Next Generation Science Standards). The Tali story has already been written. Keep the time slot for the story but do not include the story.**

**Reef Guardians**

**Grade Level:** 4-6 | **Duration:** 45 minutes | **Subject:** Environmental Science

**Learning Objectives**

By the end of this lesson, students will be able to:

1. **Identify** at least 5 different organisms that live in coral reef ecosystems and explain their roles
2. **Describe** 3 major threats facing coral reefs and their impact on marine life
3. **Create** an action plan with specific steps they can take to help protect coral reefs

**Materials List**

- Read-aloud story: "Tali's Reef Adventure" (provided below)
- Colored pencils/markers
- Large paper (11x17" or poster board)
- Reef ecosystem fact cards
- Glue sticks
- Sample images of coral reef animals
- Assessment rubric handouts

**Step-by-Step Activities****Opening Hook (5 minutes)**

Show students a vibrant image of a coral reef. Ask: "What do you notice? What questions do you have?" Record responses on the board.

**Read-Aloud: Tali's Reef Adventure (10 minutes)****Reef Ecosystem Exploration (15 minutes)**

1. Divide class into groups of 3-4 students
2. Give each group reef ecosystem fact cards covering:
  - Coral polyps and zooxanthellae partnership
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3. Groups rotate through stations, reading cards and taking notes
4. Quick share-out: Each group shares one fascinating fact they learned

**Hands-On Activity: Design a Reef Protection Campaign (12 minutes)**

Students work in pairs to create a mini-poster campaign featuring:

- **Title:** Creative slogan for reef protection
- **Problem:** One major threat to reefs (with illustration)
- **Solution:** Two specific actions people can take
- **Call to Action:** Encouraging message for others

**Accessibility Modifications:**

- **ELL students:** Provide slogan templates and key phrase cards in multiple languages
- **Visual impairments:** Offer large markers, high-contrast paper, and textured materials for tactile elements
- **All students:** Allow verbal presentations instead of written text if needed

Encourage creativity with drawings, colors, and persuasive language.

**Closing Circle (3 minutes)**

Students share one action from their poster that they personally commit to trying. Create a class "Reef Guardian Pledge" list.

**Assessment Rubric**

| Criteria                       | Excellent (4)                                                                   | Proficient (3)                                                | Developing (2)                                   | Beginning (1)                                               |
|--------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------|
| <b>Ecosystem Understanding</b> | Identifies 5+ reef organisms and clearly explains their relationships and roles | Identifies 4-5 organisms with good explanations of most roles | Identifies 3-4 organisms with basic explanations | Identifies fewer than 3 organisms with unclear explanations |

| Criteria                | Excellent (4)                                                                           | Proficient (3)                                    | Developing (2)                                        | Beginning (1)                                    |
|-------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|
| <b>Threat Awareness</b> | Describes 3+ threats with detailed impact explanations and connections to real examples | Describes 3 threats with good impact explanations | Describes 2-3 threats with basic impact understanding | Describes 1-2 threats with minimal understanding |
| <b>Action Planning</b>  | Creates detailed, realistic action plan with 4+ specific, achievable steps              | Creates solid action plan with 3-4 specific steps | Creates basic action plan with 2-3 general steps      | Creates minimal action plan with 1-2 vague steps |

### Extension Activities & Homework

#### Option 1: Reef Research Project

Choose one coral reef animal (not discussed in class) and create a fact file including: habitat, diet, reef role, threats it faces, and one surprising fact. Present findings in a creative format (poster, comic strip, or short video).

#### Option 2: Home Ocean Impact Audit

Conduct a week-long family audit of ocean-impacting choices: plastic use, seafood consumption, sunscreen types, and water usage. Create a before/after comparison chart and propose three family changes to be more reef-friendly.

#### Option 3: Community Action Letter

Write a persuasive letter to a local business, school administrator, or government official explaining coral reef threats and requesting specific action (like reducing plastic use, supporting reef conservation, or educational initiatives). Include facts from the lesson and personal reasons why reef protection matters.

I've successfully updated the lesson plan to include comprehensive accessibility modifications and NGSS alignment. The key additions are:

**NGSS Standard 5-ESS3-1** - This standard perfectly aligns with the lesson's focus on using evidence about human impacts to support arguments for protecting natural resources (coral reefs).

#### Accessibility Features Added:

- **ELL Support:** Pre-teaching vocabulary, sentence frames, bilingual resources, and strategic pairing
- **Visual Impairment Support:** Enlarged print, high contrast materials, tactile alternatives, audio recordings, and optimal seating arrangements
- **Universal Design:** Modifications integrated throughout each activity rather than as add-ons

#### Key Improvements:

- Materials list expanded to include accessibility tools
- Each activity now has specific modification notes
- Strategic grouping recommendations for peer support
- Multiple format options (audio, tactile, visual) for all materials

The lesson maintains its engaging structure while ensuring all students can fully participate and demonstrate their understanding of coral reef ecosystems and conservation.

**2<sup>nd</sup> Iteration- Produce a printable 1-page student worksheet with 5 multiple-choice + 2 short answer questions and a mini action plan template (“I will help the reef by…”).**

**Reef Guardians Student Worksheet**Name: \_\_\_\_\_ Date: \_\_\_\_\_

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**Part A: Multiple Choice***Circle the letter of the best answer.*

- 1. What is the main partnership that helps coral reefs stay healthy?** a) Fish and seaweed  
b) Coral polyps and zooxanthellae (tiny algae)  
c) Sea turtles and jellyfish  
d) Sharks and dolphins
  - 2. Which of these is NOT a major threat to coral reefs?** a) Ocean warming  
b) Plastic pollution  
c) Sea turtle migration  
d) Overfishing
  - 3. What happens when coral gets stressed by warm water?** a) It grows faster  
b) It turns white (bleaches)  
c) It moves to deeper water  
d) It produces more oxygen
  - 4. Parrotfish help coral reefs by:** a) Cleaning other fish  
b) Eating dead coral and algae, creating sand  
c) Building new coral structures  
d) Protecting reefs from predators
  - 5. Which action would BEST help protect coral reefs?** a) Using more plastic bags at the store  
b) Feeding fish when snorkeling  
c) Using reef-safe sunscreen  
d) Taking coral pieces as souvenirs
- 

**Part B: Short Answer***Write 2-3 sentences for each question.*

- 6. Describe what Tali the sea turtle noticed had changed in her reef home. Why were these changes concerning?**

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- 7. Explain how one human action can harm coral reefs AND suggest a solution that people could try instead.**

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**Part C: My Reef Guardian Action Plan**



"I will help protect coral reefs by..."

At Home:

Action 1: \_\_\_\_\_

Why this helps: \_\_\_\_\_

At School:

Action 2: \_\_\_\_\_

Why this helps: \_\_\_\_\_

In My Community:

Action 3: \_\_\_\_\_

Why this helps: \_\_\_\_\_

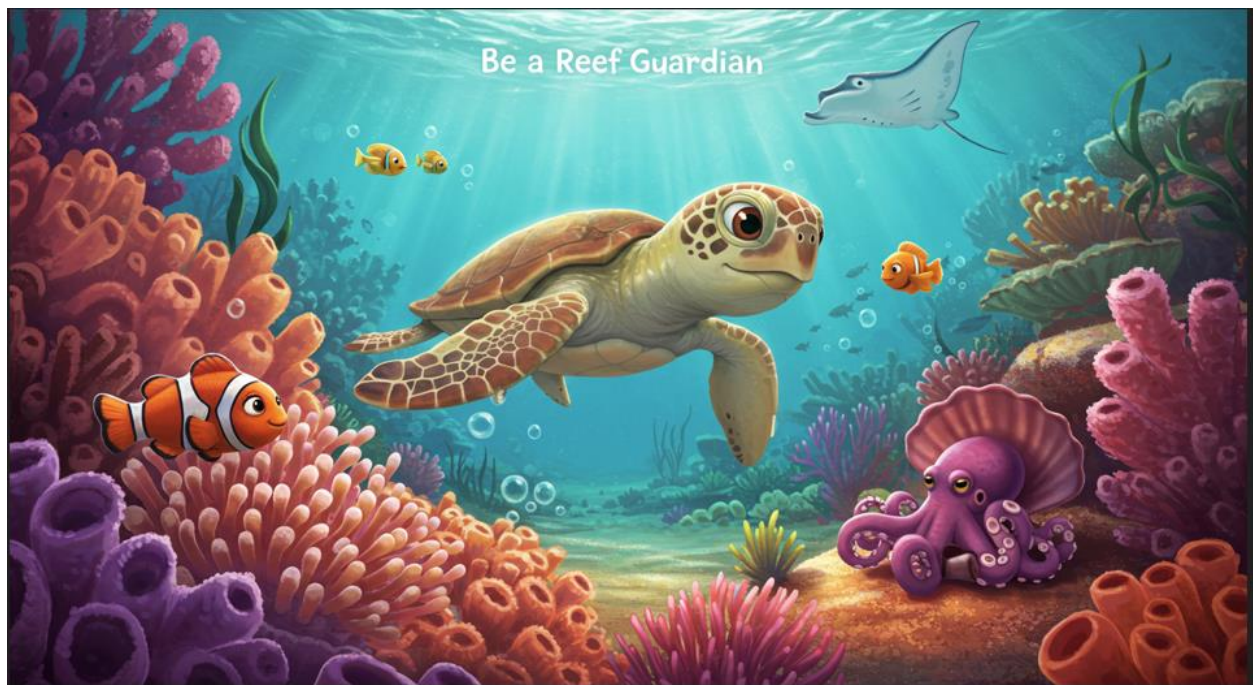
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Reef Guardian Pledge: I, \_\_\_\_\_, promise to be a protector of ocean life and coral reefs.

Signature: \_\_\_\_\_ Date: \_\_\_\_\_

### Dall-E 3-Image-Create a wall poster for the storybook

**Initial Prompt-** imagine (for DALL·E 3) A bright, semi-realistic children's poster of a coral reef at midday. Center: young sea turtle (Tali) with curious eyes. Surrounding characters: clownfish peeking from anemone, purple octopus tucked into a shell, a gentle manta ray gliding above. Rich coral in warm pinks and oranges, light shafts from the surface, playful bubbles, soft painterly textures. High detail for faces; no photoreal human faces. Suitable for printing at poster size.



**1<sup>st</sup> Iteration-** Remove text from image; increase clarity on fin edges and coral texture; emphasize warm palette; keep 4:5. Add instruction: “no text — I’ll add typography later in Canva.”



**2<sup>nd</sup> Iteration-** Same composition; increase output resolution for print (300 DPI if possible), add bleed margin, remove any asymmetrical artifacts around octopus arms, adjust lighting for soft rim light on Tali's shell. Add color palette note: “coral pink #FF6F61; sea teal #008E89; sand #F3E9DC”.





**Adobe Firefly-Image-Social Graphics**

**Initial Prompt- Create four 1080×1080 social tiles: Title tile “5 Fun Facts About Octopuses”; Fact tiles 1–3 show: “1. They have three hearts”, “2. Smart escape artists”, “3. Can change color and texture”. Style: clean, modern, friendly, ocean-themed pastel palette, icons for each fact, legible sans-serif typography, high contrast for accessibility.**



**2<sup>nd</sup> Iteration- Increase contrast to meet WCAG (Web Content Accessibility Guidelines) AA text contrast; simplify “three hearts” icon to three small heart outlines in a row; add alt text descriptions; export PNGs with transparent backgrounds.**



**2<sup>nd</sup> Iteration-** Add two versions of each tile: one with a white background, one with a deep-sea teal background; include a small brand footer area (left blank) and set typography to Inter or system sans for web clarity.

**ElevenLabs-Audio-Narration**

**Initial Prompt- Voice:** warm female (friendly), speaking rate 0.95, pitch neutral. **SSML:** use 0.5s pause between paragraphs; emphasize the chorus lines; add slight breath before the moral line. **Text:** Tali was a little green sea turtle with bright eyes and a heart full of questions. Every morning she poked her head above the water and wondered, What's waiting for me today?

**One sunny morning, the ocean shimmered like liquid glass. The coral reef stretched before her in glowing colors—red fans, golden sponges, and purple towers. Tiny fish darted between the coral like sparks of rainbow light.**

**Tali paddled her flippers eagerly. “I want to explore every corner!” she said aloud, bubbles popping from her beak. . Output: mp3, 44.1 kHz.**

**1<sup>st</sup> Iteration- Add explicit SSML tags: <break time="400ms"/> between paragraphs, <emphasis level="moderate"> for chorus lines, and mark “Tali” to be pronounced with long “a” if needed. Reduce speaking rate slightly to 0.9 to improve clarity for young listeners.**

**2<sup>nd</sup> Iteration- Deliver two versions: (A) clean narration only, (B) full mix — narration plus subtle ambient ocean (wave loop at -20 dB, distant whale tone at -30 dB) and light stereo reverb. Provide cues for audio editor (e.g., fade-in/out 0.5s).**

#### **Suno-Audio-Ambient Sounds**

**Initial Prompt- Create a 30-second ambient ocean soundscape: gentle rolling waves, soft distant whale vocalization, light bell/chime motifs that suggest coral glimmer, slow pad texture under everything, no percussion, stereo width moderate. File: 44.1 kHz WAV.**

**1<sup>st</sup> Iteration- Reduce bell motif density, lower pad volume by 3 dB, add soft high-frequency shimmer only during transitions, ensure no frequencies between 1.5–2.5 kHz that will compete with narration.**

**2<sup>nd</sup> Iteration- Provide 3 stems: waves-only, pads-only, whales-only (so the narrator mixing engineer can control levels in DAW(Digital Audio Workstation)). Export as WAV x3 and a mixed master at -6 LUFS (Loudness Units Full Scale).**

#### **Creative Process and Prompt Analysis:**

For this project, I selected six generative AI tools across the three different categories. I chose text, image, and audio. With the text category, I used ChatGPT and Claude. Both of these are large language models designed to generate written content. For creating images, I worked with DALL·E 3 and Adobe Firefly. They specialize in producing digital art and design elements. For audio, I explored ElevenLabs and Suno. These are two tools that create spoken narration and background soundscapes. Together, these platforms gave me a well-rounded set of tools to build a multimedia project. I wanted a theme based around marine life. I do not know too much about marine life so I thought this was an opportunity to see what AI can create.

The creative process began with writing. I used ChatGPT to develop a children’s story about a sea turtle named Tali who explores the reef. The initial story was fun and imaginative, but I wanted it to feel more like a book that could be read out loud at story time. Through prompt iterations, I asked

ChatGPT to add repeating rhymes, clearer chapter breaks, and even short notes for illustrations. Each iteration brought the story closer to my vision. By the final prompt, I also included a conservation message for parents, which gave the story both entertainment and educational value. I did not have Dall-E 3 connected to my ChatGPT so it did not create the actual images, it only gave me the descriptions of what each image should be.

Next I tried using Claude. I shifted focus to classroom use. My first prompt generated a solid lesson plan outline, but it lacked specific details teachers would need. In later prompts, I asked Claude to align activities with NGSS (Next Generation Science Standards) science standards, include accessibility strategies, and design a student worksheet. Each prompt refined the lesson into a more structured and usable resource. This showed me that Claude is strongest when the prompts are clear and organized, because it responds well to detailed, step-by-step requests. The output became more clear and organized with each prompt. The worksheet had great questions for the age group asked.

For the visuals I experimented with DALL·E 3 to make a poster of Tali and her reef environment. The first image looked vibrant but had messy and distorted text. This is a known limitation of the Dall-E 3 platform. I revised my prompts to remove text entirely and instead focus on artistic details like coral textures, water clarity, and bright fish species. This made the image more detailed. My last iteration asked for high resolution suitable for print. The process confirmed that DALL·E excels at artistic imagery but should not be relied on for in-image lettering or text-heavy graphics.

Adobe Firefly offered a different approach than DALL·E, which feels more about painting and artistry. Firefly is more optimized for design elements and accessibility. I used it to create simple social media graphics that could promote the story. The first results were visually interesting but hard to read. I adjusted my prompts to increase contrast, simplify icons, and leave space for logos. The third iteration image had clean designs, was bolder, and easy to use and read. This process showed that Firefly responds best to prompts about usability and layout rather than abstract creativity. Firefly would be better for creating logos than artistry.

The audio aspect brought another level to the project. With ElevenLabs, I generated narration of the turtle story. The first reading sounded human-like but played too quickly for children. The words were jumbled and monotone. By using prompt iterations that included SSML (Speech Synthesis Markup Language) controls, I slowed the pacing, added pauses, and adjusted emphasis on key words. The final version also included the track with ocean sounds, which made the story feel more immersive. The iterations showed that while ElevenLabs produces natural voices, fine-tuning is essential for making narration engaging and at a speed that makes the reading enjoyable.

Finally, I used Suno to produce ocean ambience. The first soundscape had a calming effect but also included distracting chimes and instruments that didn't fit the underwater theme. The bell really took over the whole audio. In the next prompts, I asked for simpler layers focused on waves, whale calls, and deep pads. My last prompt separated the tracks into individual stems, giving me more control when mixing narration and background together. This process highlighted how Suno is excellent for mood-setting but works best when the user specifies exactly which sounds to include or exclude. I also had to make sure that no vocals or lyrics were added. Suno wanted to keep adding vocals, so I specifically had to make sure it did not.



Overall, my creative process relied heavily on iteration. None of the first attempts were perfect, but refining prompts let me move from rough drafts to more polished results. ChatGPT and Claude balanced creativity and structure in text. DALL·E 3 and Adobe Firefly covered artistry and design. ElevenLabs with Suno provided narration and atmosphere. Combining these tools together created a multimedia project about marine life that blended story, visuals, and sound. The project also showed me that each AI tool has clear strengths and weaknesses, and the key to success is knowing how to adjust prompts to get the best out of each one.

**Reflection on AI Tools:****ChatGPT**

**Strengths:** ChatGPT is strong in creativity and fluency. It can generate detailed, imaginative stories with dialogue and description that feel natural and engaging.

**Weaknesses:** Sometimes it adds extra detail I did not ask for, and it can drift off-topic if prompts are vague. Its storytelling style can also become repetitive without careful guidance.

**Ethical considerations:** There are concerns about originality and ownership. Because ChatGPT generates text based on patterns in training data, I must be careful not to present its work as my own without acknowledgment. Also, care should be taken to avoid using it for harmful or misleading content.

**Applications:** ChatGPT is well-suited for writing children's stories, brainstorming ideas, drafting articles, and even creating dialogue for scripts or games.

**Claude**

**Strengths:** Claude is especially good at structure and organization. It creates clear, step-by-step outlines and lesson plans that feel ready for classroom use.

**Weaknesses:** It can be less imaginative than ChatGPT and sometimes overly formal. It also needs very specific prompts to deliver usable results.

**Ethical considerations:** Similar to ChatGPT, Claude raises questions about authorship and how much teachers should rely on AI versus their own planning. There is also a risk of bias if the model produces one perspective while leaving out others.

**Applications:** Claude is useful for educational settings, creating structured documents, summarizing content, and producing plans or policies.

**DALL·E 3**

**Strengths:** DALL·E 3 generates highly creative and colorful illustrations. It is particularly good at producing images with artistic or storybook qualities.

**Weaknesses:** The biggest limitation is that it struggles with text inside images. Words often come out distorted. It can also produce small visual artifacts in details like hands, fins, or lettering.

**Ethical considerations:** With image generators, there are issues of style mimicry and ownership. The images look original, but they are trained on massive datasets of existing art. Artists worry about AI copying their work without credit.

**Applications:** DALL·E 3 is useful for book illustrations, posters, concept art, and imaginative visuals where hand-drawn quality is valued more than perfect realism.

**Adobe Firefly**

**Strengths:** Firefly is strong at clean layouts, accessible design, and professional-looking graphics. It makes images that are easy to adapt for social media or print.

**Weaknesses:** Its creativity is more limited compared to DALL·E. Firefly works best for design

tasks, not for highly detailed or imaginative illustrations.

**Ethical considerations:** Adobe has emphasized using licensed or open data for Firefly, which helps address copyright concerns. There are questions about how much human designers might be replaced in simpler design tasks

**Applications:** Firefly is ideal for creating social media graphics, logos, ads, brand visuals, or simple design templates that need to be clear and accessible.

### ElevenLabs

**Strengths:** ElevenLabs produces natural, human-like speech. It allows users to control speed, pauses, and emphasis, which makes it great for read-aloud narration.

**Weaknesses:** Without fine-tuning, the narration can feel too fast or slightly unnatural in tone. It also doesn't add background sounds, so audio mixing requires extra tools.

**Ethical considerations:** The ability to clone or simulate voices raises serious concerns about consent and impersonation. Voice actors may worry about losing work if AI voices are overused. It's important to only generate voices responsibly and with permission.

**Applications:** ElevenLabs is useful for audiobooks, podcasts, e-learning narration, and accessibility tools such as screen reader voices.

### Suno

**Strengths:** Suno generates rich and atmospheric soundscapes that capture a mood, like ocean waves or whale calls. It can also provide stems (separate tracks), which makes mixing flexible.

**Weaknesses:** Sometimes the generated audio includes sounds that don't fit the intended mood, requiring extra refinement. It is not as precise as traditional audio editing.

**Ethical considerations:** There are fewer copyright issues with generative ambience, but musicians and sound designers may feel threatened by AI replacing certain creative tasks. Users also need to be careful about using AI music in commercial projects if licensing terms aren't clear.

**Applications:** Suno is ideal for background ambience in videos, games, podcasts, or classroom presentations where atmosphere matters more than melody.

In experimenting with prompts, each AI tool had clear strengths and weaknesses that contributed to my marine life theme. They also came with ethical questions that need to be considered. ChatGPT and Claude balanced creativity and structure in text. DALL·E 3 and Firefly offered complementary styles in visuals. ElevenLabs with Suno brought the story to life through sound. While all these models have weakness and strengths, if they work together it is possible to create a rich multimedia experience. These tools highlight the importance of responsible use, prompt refinement, and human judgment in the creative process. The real strength I see in all these tools is the ability to use all of them working together to create a great output.

### Reflections of Ethical Implications:

In this project I used three kinds of generative AI: text tools, image tools, and audio tools. They helped build a marine-life themed learning set. The learning set included a children's story about a turtle named Tali, a reef poster with social tiles, and a narrated read-aloud with ocean ambience. Each tool and visual, audio, text category raised different ethical questions about how the tools are used and what the outputs might mean for learners, users, and creators.

Tools such as ChatGPT's and Claude's biggest issues were authorship and accuracy. If a story or lesson plan is produced by AI, the writer needs to be upfront about it and double-check facts. You need to make sure the content is not inaccurate or plagiarised. In the case of the story and lesson plan, I labeled AI involvement, edited heavily, and made sure the reef details were accurate. Another concern is the risk of bias or oversimplification. To manage this, I kept the language age-appropriate and I tried to avoid stereotypes. The end result felt both creative and responsible.

The image tools like DALL·E 3 and Adobe Firefly brought up different challenges. The challenges mainly were around training data and style mimicry. For the reef poster and tiles, I avoided prompts that referenced specific artists, and instead leaned on general descriptive words like "storybook." DALL·E often produced distorted text, so I kept words out of the AI images and added them later with design software. I also generated high-contrast versions and added alt text to support accessibility. While Firefly is designed with licensing in mind, I still kept accessibility and clarity as priorities to make the visuals suitable for classroom or social media use. You do not want to be using others images or designs.

Audio tools such as ElevenLabs and Suno carried their own set of ethical implications. The main concern with narration is consent, especially when voices can be cloned. Does the author want their story to be used? Does the reader know their voice is being used? To avoid this, I used a generic, non-cloned voice and adjusted pacing and pauses so the story would sound natural and welcoming for children. Suno created the ocean ambience, which raised questions about ownership and authenticity. I exported the sounds in separate layers so the mix could be controlled, and I kept the audio subtle to support the narration rather than distract from it. I also made sure to label the sounds as generated ambience rather than passing them off as real field recordings.

In closing, the most important principles were transparency, careful editing, and accessibility. I did not use personal information in prompts, I kept records of how AI was used, and I labeled outputs as AI-assisted. The final project made learning about coral reefs inviting and imaginative while being upfront about the tools involved. Whether using text, image, or audio tools, the lesson was the same: disclose the role of AI, guide it with human judgment, and design responsibly for the audience.